

Week 1 · Task 1 – Cyber Security & Ethical Hacking – Introduction

1. The CIA Triad

The CIA Triad is the foundational model designed to guide information security policies within an organization. It balances three core security concepts to ensure system resilience and data protection:

- **Confidentiality:** Ensuring that sensitive information is accessible only to authorized personnel and protected from unauthorized disclosure.

Real-World Example: Implementing Multi-Factor Authentication (MFA) and AES-256 encryption on corporate email servers to prevent unauthorized parties from viewing financial reports.

- **Integrity:** Maintaining the accuracy, completeness, and trustworthiness of data by protecting it against unauthorized modification or tampering.

Real-World Example: Utilizing SHA-256 cryptographic hashes and digital signatures during file transfers to verify that software update packages have not been modified by a Man-in-the-Middle (MitM) attacker.

- **Availability:** Ensuring that systems, applications, and data are timely accessible to authorized users whenever needed.

Real-World Example: Deploying redundant server clusters, load balancers, and Cloudflare DDoS mitigation services to keep e-commerce sites operational during high-volume traffic or cyber attacks.

2. Hacker Classification: Black, White, and Grey Hat

Threat actors and security professionals are broadly categorized based on their intent, authorization, and adherence to legal frameworks:

Hacker Type	Primary Intent	Authorization	Operational Context
Black Hat	Malicious gain, financial theft, sabotage, or espionage.	Unauthorized (Illegal)	Operates in secret to exploit vulnerabilities, deploy ransomware, or steal sensitive data.
White Hat	Defensive security, identifying flaws to fix them.	Fully Authorized (Legal)	Employed as security analysts, penetration testers, or bug bounty hunters under explicit consent.

Grey Hat	Curiosity or exposure, though without malicious intent.	Unauthorized / Ambiguous	Finds vulnerabilities without prior permission and notifies the vendor, sometimes requesting a disclosure fee.
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3. The 5 Phases of Ethical Hacking

In a structured penetration testing engagement, ethical hackers follow a standardized 5-phase methodology to evaluate target environments:

- **1. Reconnaissance (Footprinting):** Gathering preliminary information and intelligence about the target system or organization prior to launching an evaluation.
- **2. Scanning & Enumeration:** Identifying active hosts, open network ports, running services, and live vulnerabilities using technical scanning tools.
- **3. Gaining Access (Exploitation):** Exploiting discovered vulnerabilities to enter the target system and execute unauthorized commands or access data.
- **4. Maintaining Access:** Establishing persistent access mechanisms, such as backdoors or elevated privileges, to assess long-term exposure.
- **5. Covering Tracks & Reporting:** Removing diagnostic artifacts and log traces, followed by delivering a technical remediation report to stakeholders.

4. Vulnerability Assessment vs. Penetration Testing

While both practices aim to enhance security posture, they differ significantly in scope, depth, and methodology:

Feature / Metric	Vulnerability Assessment (VA)	Penetration Testing (PT)
Primary Objective	Identify, list, and prioritize known vulnerabilities across assets.	Demonstrate the real-world impact of exploiting security flaws.
Approach & Depth	Automated, broad-scoped scan across network/systems.	Manual & targeted exploitation probing specific attack paths.
Frequency	Regular / Continuous (e.g., weekly, monthly).	Periodic / Event-driven (e.g., annual, major releases).

Output / Deliverable	Prioritized list of unpatched flaws & CVE ratings.	Proof of Concept (PoC), exploit chains, & root-cause analysis.
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Risk Level	Low (non-disruptive scanning).	Moderate to High (simulated live attacks require caution).
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